ATLAS: Network Atlas, a hyperbolic network exhibit

Design and plan, version 1, 4 October 2026. Proposal for the lead; not part of the Master Plan yet.

Based on: D. Krioukov, F. Papadopoulos, M. Kitsak, A. Vahdat, M. Boguñá, *Hyperbolic Geometry of Complex Networks*, Phys. Rev. E 82, 036106 (2010), arXiv:1006.5169. Prototype and figures: `hypernet.py` in this folder.

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0. Summary

What. An exhibit that shows a real-looking network (an "Internet" of 200 nodes) living in the hyperbolic plane. The visitor sees:

- why the network has a few huge hubs and many small nodes;
- why a message finds its way with no map, just by stepping "closer" each time.

Why it fits ATLAS:

- ATLAS already has a hyperbolic kernel, a Poincaré-disc map and a greedy beacon. The Lung airway uses the same idea: a tree fits in hyperbolic space.
- The exhibit reuses those parts and adds a second "beyond gaming" story next to the lung.

Three views, cheapest first:

View	Where	What the visitor does	New code
V1 Station disc	Console web page and Spectator	Watches the network in the Poincaré disc; a greedy route animates hop by hop	Small: Python generator + one JS page
V2 Floor map	Headset, in Home, standing	A 2.4 m Poincaré disc lies on the floor; looking at a node re-centres the world on it (focus + context)	Medium: one blade, flat meshes
V3 Walk the network	Headset, in Curved Walk (S2)	Stands at a node in true hyperbolic space; beacons point along the real geodesics to its neighbours; "hops" carry the packet	Large: tile placement, beacons, hops

Recommendation:

- Build V1, then V2.
- Keep V3 as a stretch goal, because a whole network does not fit a room (section 5).
- Start only after gate G7, or earlier only with the lead's agreement: the Master Plan allows only fixes until G1 to G6 are green (§12.3).
- V1 touches only the Station, so it can be prototyped without touching the phone app.

Measured on the prototype (N = 200, mean degree 5.8, γ = 2.5, T = 0, seed 7, 2,000 random pairs):

Greedy routing using	Success	Mean hop stretch
Hyperbolic distance	**100 %**	**1.00** (every route as short as the shortest path)
Straight-line distance in the Poincaré picture	74 %	1.03

This is the headline the exhibit can show live: the same network, the same rule ("step closer"), and only the geometry changes.

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1. Why hyperbolic geometry for networks

Fact	Flat plane	Hyperbolic plane
Circumference of a circle of radius r	$2\pi r$ (grows linearly)	$2\pi \sinh r \approx \pi e^r$ (grows exponentially)
Nodes of a tree at depth r, b children each	b^r (exponential)	b^r fits without crowding

Real networks are roughly tree-like. ISPs sit above regional networks, which sit above local ones; communities sit above sub-communities. A tree needs exponentially growing room, which only hyperbolic space has.

The paper places nodes in a hyperbolic disc and links the ones that are close. From that one rule, three things follow:

1. **Hubs.** Nodes near the centre are close to almost everyone, so they get many links. The degree distribution is a power law, $P(k) \sim k^{(-\gamma)}$, with $\gamma = 2\alpha/\zeta + 1$ (eq. 29).
2. **Clustering.** Distance obeys the triangle inequality, so friends of friends are linked and triangles form.
3. **Navigation without a map.** Greedy forwarding (pass to the neighbour closest to the target) succeeds about 99.9 % of the time on Internet-like networks (§X.1) and survives the loss of 30 % of the links.

Radius means **popularity**; angle means **similarity**. Shortest paths dip towards the centre and back out, like going up to a common ancestor in a tree.

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2. What the visitor should take away

Moment	The visitor sees	The "aha"
Hubs	Three or four big nodes near the centre, most nodes on the rim	"The rich are in the middle and connected to everyone"
Room at the edge	Re-centre on any node: its neighbourhood opens up and the rest shrinks to the rim	"There is always room for more branches"
Two Compasses for networks	The same packet delivered twice: hyperbolic greedy arrives; flat-picture greedy gets stuck	"The geometry is what makes the map-free rule work"
Live number	Success rate and stretch over many random packets	Every claim backed by a number (rubric: geometric creativity 30 %, correct maths)

Comparison with the rubric:

- **Geometric creativity:** the geometry is not decoration; it explains the network.
- **Educational value and "beyond gaming":** this is how the Internet and social networks are actually studied.
- **Documentation:** one maths card per moment.

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3. The model

All formulas use $\zeta = 1$, so one unit of hyperbolic distance is one curvature radius. In ATLAS that unit is **L metres**, the room's loop-fit: L5 = 1.88 m in `sample_3x3`.

Quantity	Formula	Source
Exponent of the radial density	$\alpha = (\gamma - 1) / 2$	eq. 29 with $\zeta = 1$
Helper	$\xi = \alpha / (\alpha - 1/2)$	eq. 25

Quantity	Formula	Source
Scale	$\nu = \pi \bar{k} / (2 \xi^2)$ at $T = 0$; $\nu = \bar{k} \sin(\pi T) / (2 T \xi^2)$ for $0 < T < 1$	eqs. 25, 31; hot case of §VII
Disc radius	$R = 2 \ln(N / \nu)$; for $\gamma = 3$ this is $R = 2 \ln(8N / (\pi \bar{k}))$	eq. 13
Radius of a node	$r = \operatorname{arccosh}(1 + u (\cosh \alpha R - 1)) / \alpha$, with u uniform in $[0, 1]$	exact inverse of the CDF of eq. 17
Angle	θ uniform in $[0, 2\pi)$	
Distance	$\cosh x = \cosh r \cosh r' - \sinh r \sinh r' \cos \Delta\theta$, with $\Delta\theta = \pi - \lvert\pi - \lvert\theta - \theta'\rvert\rvert$	eq. 5
Link, $T = 0$	linked if $x < R$	eq. 8
Link, $T > 0$	$p(x) = 1 / (e^{((x - R) / (2T))} + 1)$	eq. 41
Greedy forwarding	next = the neighbour with the smallest x to the target; drop when no neighbour is closer	§X
Poincaré disc point	$\rho = \tanh(r / 2)$, at angle θ	conformal model
Re-centring on point a	$z \to (z - a) / (1 - \bar{a} z)$	Möbius isometry of the disc
In the ATLAS kernel frame	$P = (\sinh r \cos\theta, \sinh r \sin\theta, \cosh r)$, metric $\operatorname{diag}(1, 1, -1)$, $\kappa = -1$	lane B contract; `Geometry2.Distance`

Default parameters: $N = 200$, $\bar{k} = 6$, $\gamma = 2.5$ (the Internet is about 2.1), $T = 0$ (the hot case adds randomness). These give $R = 10.5$ units.

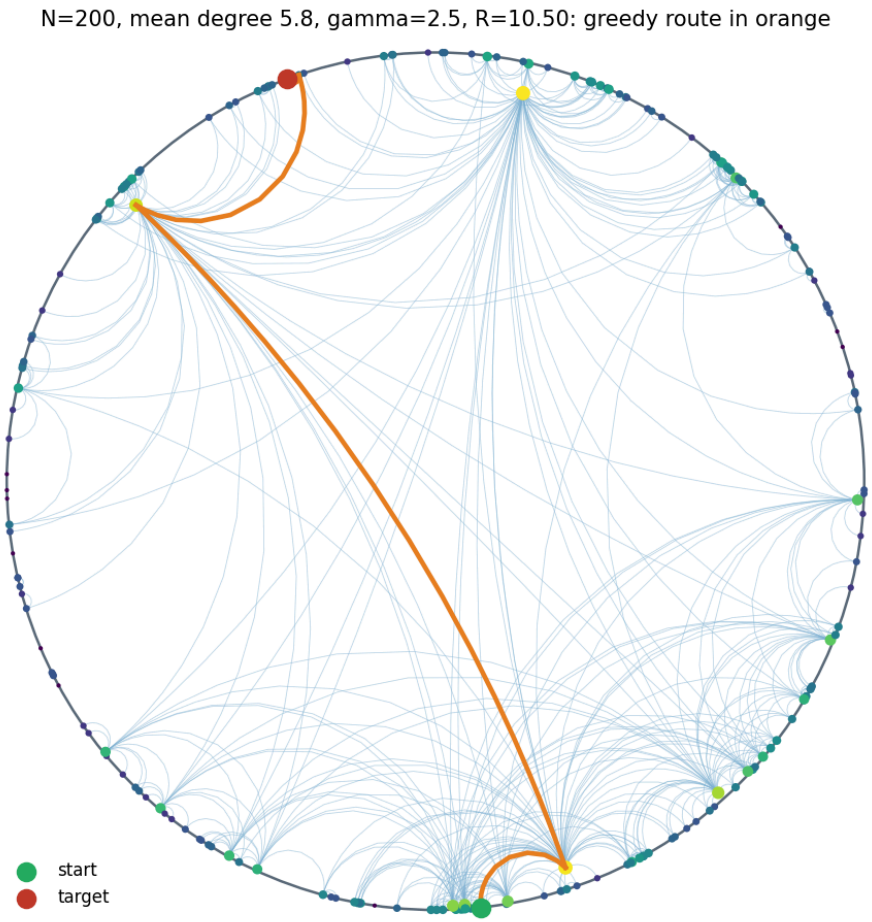


Figure 1. The prototype network in the Poincaré disc. Edges are drawn as true geodesics. Node size and colour follow degree. The orange line is a greedy route: from a rim node, through two hubs, to a far rim node, in 4 hops.

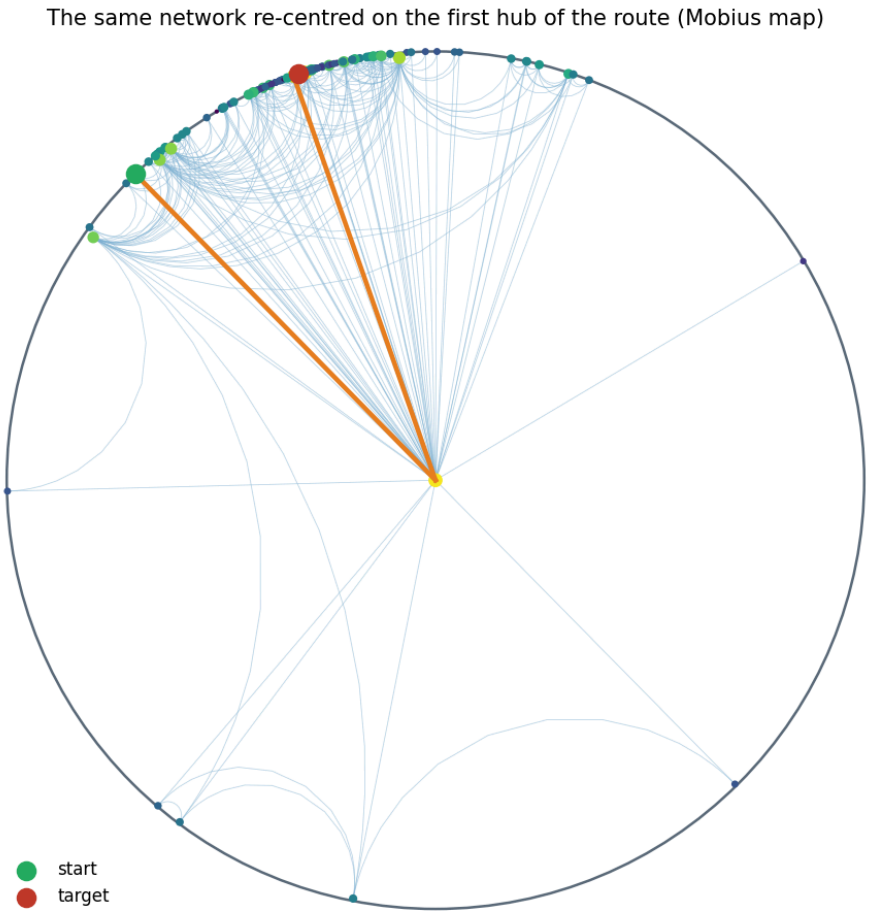


Figure 2. The same network re-centred on the first hub of the route by a Möbius isometry. Distances are unchanged; only the point of view moves. This is the V2 interaction.

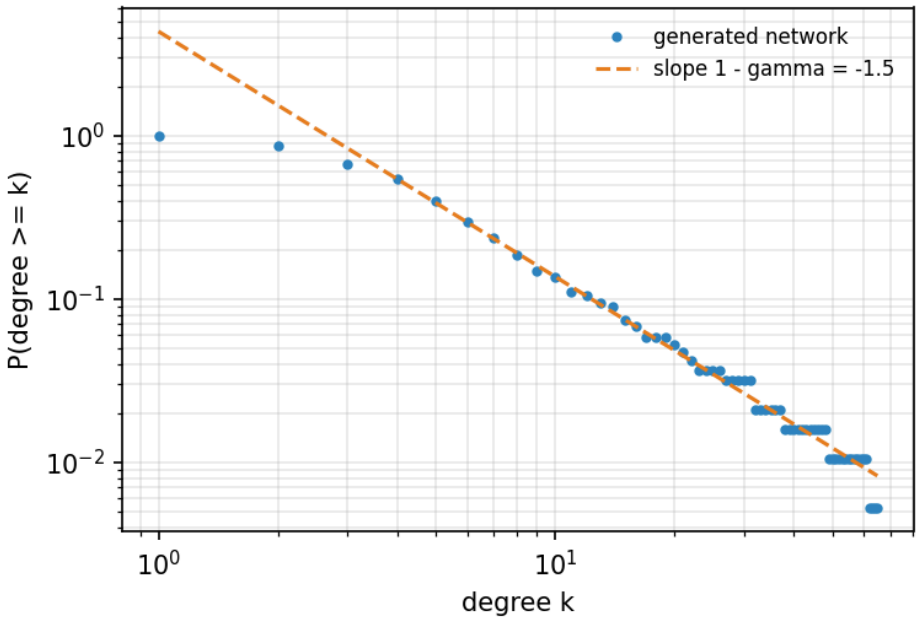


Figure 3. The share of nodes with degree ≥ k on log-log axes. The tail follows the power law of slope 1 − γ (one hub has degree 65; the mean is 5.8).

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4. The three views

V1: Station disc (Console and Spectator)

Experience:

- A Console page draws the network in the Poincaré disc.
- The operator picks two nodes, or presses *Random packet*.
- The route animates hop by hop: first with the hyperbolic rule, then with the flat-picture rule.
- A panel shows success rate and stretch over 1,000 random packets for both rules.
- The Spectator shows the same disc, so the audience can follow along.

How:

- `station/pipelines/atlas_pipelines/network.py` generates the network. It ports `hypernet.py` into the existing `atlas_pipelines` package, next to `airway.py`.
- It writes `network_<seed>.json` (section 6.3) and runs the routing study offline.
- `station/hub/wwwroot/js/network.js` draws geodesics as sampled arcs (the method of `hypernet.geodesic_disk`) on a canvas, in the style of `mapview.js`.

Reuses: the Hub's static file serving, the Console layout and `mapview.js` drawing helpers; on the Spectator, `Telemetry/Map/FlatTwinDisc.cs` (the Poincaré disc already drawn for S2).

Cost: no phone code and no frame budget. This is the lowest-risk way to show the idea to judges.

V2: Floor map in the headset (standing)

Experience:

- In Home, a disc 2.4 m across lies on the floor, inside the free rectangle. It is the Poincaré disc of the network.
- Nodes are small posts whose height grows with log(degree); edges are thin geodesic arcs on the floor.
- Look at a node for 0.7 s (or point the gamepad at it and press A): the whole map glides so that node comes to the centre (a Möbius map over 0.6 s).
- Its neighbourhood opens up around the visitor's feet while the rest of the world shrinks to the rim. This is "focus + context" (Lamping and Rao's hyperbolic browser), and it shows why hubs are close to everything.
- *Send packet*: pick a target, and the route lights up hop by hop, with the hop count and the stretch on a caption chip.

Why standing:

- The visitor never walks in curved space, so comfort is not an issue.
- It works in the smallest rooms, including standing mode.
- It works before gate G2, the curved-walk comfort test, has passed.

How:

- The core is engine-free, in `Blades/Network/Core/`:
  - `NetworkModel` (load the JSON, distances);
  - `PoincareMap` (project, Möbius re-centre, ease between two centres);
  - `GreedyRouter`.
- The views use the look:
  - `NetworkFloorView` builds node posts and edge strips with `AtlasMeshBuilder`, colours from `AtlasPalette`, materials from `MaterialPool`;
  - the mesh is rebuilt only while a re-centre animates, in at most 3 merged meshes (nodes, edges, highlight).
- Twins:
  - gaze dwell on node targets registered in `GazeRegistry` (at most 64 at once: the nodes nearest the centre after projection);

- gamepad: stick to move a cursor between nodes, A to re-centre, Y to send a packet.

Budget:

- 200 nodes × 1 post = 200 small boxes (about 2.4k triangles).
- 600 edges × 16 segments × 2 triangles ≈ 19k triangles.
- 3 draws per eye. Well inside the S0 row of §8.5.

V3: Walk the network (Curved Walk, stretch goal)

Experience:

- The visitor stands *at* a node, in true hyperbolic space (S2, Mode 1), on its own tile.
- Its neighbours are shown as **beacons**: arrows on a ring around the visitor, each pointing along the true geodesic. The direction is the kernel's log map, as in the Lung's `AirwayBeacon`.
- The labels give each neighbour's distance and degree.
- Nodes within the fog radius (ρ = 3 units ≈ 5.6 m) are drawn as towers in their tiles; the visitor can walk to them.
- To follow an edge longer than the room allows, the visitor **hops**: dwell on the beacon, the world fades to black for 0.25 s, the kernel re-roots at the neighbour node, and the world fades back in.
- The packet game: deliver a packet to a target using only the beacons. Afterwards the Console shows the visitor's route beside greedy and shortest.

How:

- Node placement: the point P of each node (section 3) is snapped to the {4,5} tile graph by stepping from tile zero towards P (`TileGraph.Neighbour`, choosing the neighbour whose centre is nearest P). The node's offset inside that tile goes into the tile chart, as the airway is placed in `AirwayTreeBinding`.
- Hosting follows the Lung pattern: `NetworkWingHost` is registered with Curved Walk like `AirwayWingHost`.
- Hops reuse SpaceLab's fade and the kernel's re-root (`HolonomyWalkerKernel`), the same machinery as *Return Home*.

Why it is a stretch goal: see section 5.

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5. The scale problem: why the whole network cannot be walked

In ATLAS one hyperbolic unit is L metres (1.88 m in a 3 × 3 m room). The disc radius grows only **logarithmically** with N, so even small networks are far too wide to walk:

Network	R (units)	R in a 3 × 3 m room	Rim to rim through the centre
N = 200, $\bar{k}$ = 6	10.5	19.8 m	about 40 m
N = 100, $\bar{k}$ = 6	9.1	17.2 m	about 34 m
N = 60, $\bar{k}$ = 5	8.5	15.9 m	about 32 m
N = 30, $\bar{k}$ = 4	7.5	14.2 m	about 28 m

Further facts:

- A link can be up to R long, so most neighbours sit beyond the fog (5.6 m).
- The greedy route in figure 1 covers 20.9 units, about 39 m of walking.
- Shrinking the units is not allowed. L is fixed by the room and by the comfort floor (L ≥ 1.4 m), and gains are not allowed (real walking only, Master Plan §10.1.4).

Consequence:

- V1 and V2 show the whole network at once, which is what the paper is about.
- V3 can only show the local view, and it needs hops.
- That is honest and still striking ("this is what being a router feels like"), but it is more work for less overview.

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6. Architecture

6.1 Data flow

```
Station PC                                      Phone
----------                                      -----
atlas_pipelines/network.py                      StreamingAssets/networks/network_<seed>.json
  generate (N, k̄, γ, T, seed)       --bundle-->   (built in; no Station needed)
  greedy study (hyperbolic vs flat)                 |
  write network_<seed>.json                         v
       |                                        Blades/Network/Core  (engine-free)
       v                                          NetworkModel, PoincareMap, GreedyRouter
Hub /api/network  ----------------------------->    |
Console network.js (V1)                             v
Spectator FlatTwinDisc (V1)                     Blades/Network  (views)
                                                  V2 NetworkFloorView in Home
       ^                                          V3 NetworkWingHost in Curved Walk
       |  events net.* and State
       +----------------------------------------  LinkClient (unchanged)
```

6.2 Code layout

Path	What	Lane
`station/pipelines/atlas_pipelines/network.py` + `tests/test_network.py`	Generator, greedy study, JSON writer (port of `hypernet.py`)	D
`station/hub/wwwroot/js/network.js`, Console tab "Network"	V1 drawing and packet animation	D
`ATLAS/Assets/Atlas/Blades/Network/Atlas.Blades.Network.asmdef`	New blade assembly; references `Atlas.World`, `Atlas.Kernel`, `Atlas.Rendering`, `Atlas.Input`, `Unity.Mathematics`	A or C
`Blades/Network/Core/NetworkModel.cs`, `PoincareMap.cs`, `GreedyRouter.cs`	Engine-free core (no `UnityEngine` type)	A or C
`Blades/Network/NetworkFloorView.cs`, `NetworkTwins.cs`	V2 meshes and gaze targets	A or C
`Blades/Network/Walk/NetworkWingHost.cs`, `NetworkBeaconView.cs`	V3 (stretch)	B
`Tests/Network/*`	EditMode tests, run by `tools/headless/verify.py` too	all

Rules this design keeps (contracts F1 to F9; verification brief §1):

- Blades never reference Scan, Safety or App (F1). The blade plugs in through `ISpaceModule` and the Lung's host pattern.
- The core is engine-free (F2).
- Everything is built in code (F3). The JSON is data, not a scene.
- One shader, and no `MaterialPropertyBlock` (F7): meshes use the shared `AtlasVertex` layout and `MaterialPool`.

- No allocation per frame on the frame path. Meshes are rebuilt only during a re-centre, into preallocated buffers.
- Bump Guard is untouched and keeps working in every view.

6.3 Data file `network.json`

```
{
  "schema": "atlas.network/1",
  "model": { "n": 200, "kbar": 6, "gamma": 2.5, "T": 0, "seed": 7, "zeta": 1, "R": 10.504 },
  "nodes": [ { "id": 0, "r": 9.84, "theta": 1.23, "degree": 4, "label": "AS-0" } ],
  "edges": [ [0, 17], [0, 42] ],
  "study": {
    "pairs": 2000,
    "hyperbolic": { "success": 1.0, "stretch": 1.0 },
    "flat_picture": { "success": 0.741, "stretch": 1.026 }
  }
}
```

6.4 Parameters, events and twins

Kind	Name	Values	Purpose
Parameter	`net.file`	enum of bundled networks	Which network is shown
Parameter	`net.view`	`floor`, `walk`	V2 or V3
Parameter	`net.rule`	`hyperbolic`, `flat`	Routing rule shown
Parameter	`net.recentreSeconds`	0.3 to 1.5 (0.6)	Re-centre animation length
Parameter	`net.maxGazeNodes`	16 to 64 (64)	Gaze targets registered at once
Event	`net.recentre`	`{node, r, degree}`	Each re-centre
Event	`net.packet`	`{src, dst, rule, hops, shortest, stretch, success}`	Each delivered or dropped packet
Event	`net.hop` (V3)	`{from, to, distance}`	Each hop
Twins	Re-centre	gaze dwell on the node / stick cursor + A	Rule 5 of the Master Plan
Twins	Send packet	gaze dwell on the *packet* chip / Y	
Twins	Switch rule	gaze dwell on the *rule* chip / X	

Parameters are declared with `[AtlasParam]`, so the Console tuner shows them with no extra Console code.

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7. Plan

The same style as the Master Plan: each task fits one coding agent, touches one lane's files, and has a binary acceptance test with numbers.

ID	Task	Lane	Depends on	Output	Acceptance (binary)	Needs
NET-01	Network generator and study	D	none	`atlas_pipelines/network.py`, pytest	Mean degree within 10 % of $\bar{k}$ for N = 200 over 10 seeds; CCDF tail slope within 0.3 of 1 − γ; hyperbolic greedy success ≥ 99 % and flat-picture success ≤ 90 % over 2,000 pairs; output validates against `atlas.network/1`	Station
NET-02	Network contract	D	NET-01	`docs/contracts/network.md`, JSON schema	Schema checked in CI; two seeds bundled under `StreamingAssets/networks/`	Editor
NET-03	V1 Console page	D	NET-01	`network.js`, Console tab, `/api/network`	Draws 200 nodes and 580 edges in under 50 ms in a real browser; a packet animates hop by hop; node tests (`node --test`) green	Station
NET-04	V1 on the Spectator	D	NET-02, P1-D07	Disc in the Spectator's map panel	Same picture as the Console for the same seed (node positions equal to 1 px); ≤ 0.3 ms per frame	Station
NET-05	Engine-free core	A or C	NET-02	`NetworkModel`, `PoincareMap`, `GreedyRouter` + EditMode tests	Distances equal the Python ones to 1e-9; a Möbius re-centre preserves every pairwise hyperbolic distance to 1e-9; greedy routes equal the Python routes on 1,000 pairs	Editor
NET-06	V2 floor map	A or C	NET-05	`NetworkFloorView`, blade registration in Home	3 draws per eye; ≤ 25k triangles; 0 B managed allocation per frame at rest; re-centre ≤ 2 ms CPU per frame	Editor, then phone in headset
NET-07	V2 twins and packet	C	NET-06	`NetworkTwins`, caption chip, `net.*` events	Every action by gamepad alone and by gaze alone; `net.packet` logged for 10 of 10 packets with hops and stretch equal to the core's	Phone in headset
NET-08	Exhibit wiring	C	NET-07, P2-C08	Exhibit sign in Home	Reachable from the exhibit menu; returns Home with the room unchanged (walls within 0.05 m)	Phone in headset
NET-09	V3 node placement (stretch)	B	NET-05, P1-B04	Snapping to tiles, towers in tiles	Every node's tile is the tile whose centre is nearest (checked against brute force on 200 nodes); WorldLint passes	Editor
NET-10	V3 beacons and hops (stretch)	B	NET-09	`NetworkWingHost`, `NetworkBeaconView`	Beacon bearing within 1° of the log map; a hop re-roots within 0.5 s with a fade and no visible pop	Phone in headset
NET-11	Maths card and docs	B	NET-01	Section in `docs/maths.md`, deck slide	Every formula of section 3 names a passing test; no claim beyond section 8	Editor

Milestones:

Milestone	Tasks	What can be shown
N1 Station disc	NET-01 to NET-04, NET-11	Judges watch hyperbolic vs flat routing on the Console and Spectator
N2 Floor map	NET-05 to NET-08	Visitor re-centres the network around them in the headset
N3 Walk (stretch)	NET-09, NET-10	Visitor routes a packet from inside hyperbolic space

Where it sits in the Master Plan:

- It is not in the plan today. The lead would add it as a Tier B blade, next to the Playground and Lung tracks of Phase 2.
- **N1 and N2 should start only after G7.** The plan allows only fixes until G1 to G6 are green (§12.3).
- N1 lives entirely on the Station, so lane D can prototype it earlier without touching the phone.
- N3 depends on G2: the curved-walk comfort result decides whether visitors walk in S2 at all.

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8. Claims we make and claims we avoid

We say	We do not say
"This network was *generated* by the hyperbolic model of Krioukov et al. 2010; real networks share its hubs, clustering and navigability."	"This is the real Internet" (unless a real embedded map is used, and credited)
"With the hyperbolic rule, greedy delivery succeeded on X of Y random packets in *this* network."	"Greedy routing always works" (it is measured, not guaranteed)
"Flat-picture greedy fails more often on the same network."	"Euclidean geometry cannot route" (a different flat embedding could do better)
"Hyperbolic space has room for every branch of a tree."	Anything clinical about the Lung by analogy

This follows the Master Plan's risk 19 (over-claiming the hyperbolic airway): numbers are measured, not assumed.

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9. Risks

Risk	Effect	Mitigation
Phase 1 not finished	Exhibit work steals time from the gates	Start after G7; N1 only on the Station
Clutter: 580 edges look like noise	The visitor sees a hairball	Show edges of the focus node and the route by default; all edges on a toggle; fade edges by distance from the centre
Nodes pile up on the rim ($\tanh(r/2) \approx 1$)	Rim nodes are hard to pick by gaze	Gaze targets only for the 64 nodes nearest the centre after re-centring; re-centre to reach the others
V3 too long to walk	Visitor fatigue, sickness	Hops with a fade; V3 stays a stretch goal; gated on G2
Real Internet data licence	Cannot ship real data	Synthetic networks first; real data only with a cleared licence (open question 2)

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10. Open questions for the lead

1. Add the exhibit to the plan at all, and as which tier: B (demo) or C (roadmap)?
2. Real data:

- Boguñá, Papadopoulos and Krioukov published a hyperbolic map of the Internet (*Sustaining the Internet with hyperbolic mapping*, Nature Communications, 2010). Is a real embedding worth the licence check?
- Embedding any graph, for example the team's own code-dependency graph, needs an embedding tool (for example Mercator). Is that in scope?

3. Owner for V2: lane C (exhibits) or lane A (look)?
4. Which γ to show: 2.1 (Internet-like, a few giant hubs) or 2.5 (calmer picture)?

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11. References

- D. Krioukov, F. Papadopoulos, M. Kitsak, A. Vahdat, M. Boguñá. *Hyperbolic Geometry of Complex Networks*. Phys. Rev. E 82, 036106 (2010). arXiv:1006.5169.
- M. Boguñá, F. Papadopoulos, D. Krioukov. *Sustaining the Internet with hyperbolic mapping*. Nature Communications 1, 62 (2010).
- R. Kleinberg. *Geographic routing using hyperbolic space*. INFOCOM 2007 (every tree has a greedy embedding in H^2; already cited by the ATLAS plan for Two Compasses).
- J. Lamping, R. Rao. *The hyperbolic browser: a focus + context technique for visualizing large hierarchies*. Journal of Visual Languages and Computing 7 (1996).
- ATLAS Master Plan: §3 (spaces), §4.6 to §4.8 (Curved Walk and Aligned Cells), §8.5 (budgets), §10.1.4 (comfort rules), §10.2 (Lung airway, Two Compasses), §12.3 (if a gate fails).

Appendix: reproduce the numbers

```
python hypernet.py --n 200 --kbar 6 --gamma 2.5 --T 0 --seed 7 --pairs 2000
```

Writes `figures/fig1_poincare_route.png`, `fig2_recentred.png`, `fig3_degrees.png` and `stats.json`:

- R = 10.504; mean degree 5.8; maximum degree 65;
- giant component 179 of 200 nodes; 580 edges;
- greedy hyperbolic: 100 % success, stretch 1.00;
- greedy flat-picture: 74.1 % success, stretch 1.03.